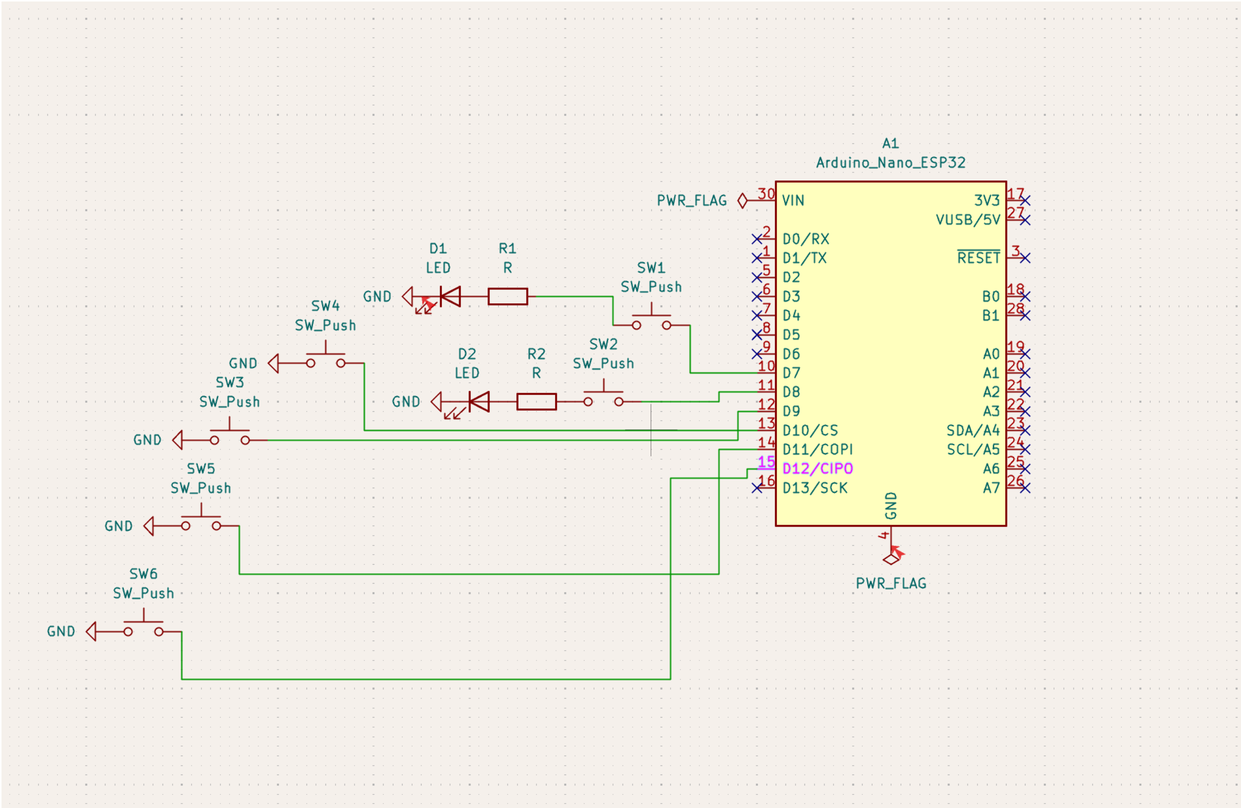
GHardware Final Part 2 Explanation

For this part, I worked on Excite Bike. I designed my controller with the given constraints in mind, four directional inputs and two buttons, on for throttle and one for turbo boost. I considered the comfortability of the design, so I made sure that the buttons where a good distance from each other and not too difficult to reach. For Aesthetics, I decided to give the controller’s handles cylindrical shapes to represent a bike’s handles, as I thought this was a clever way to be on theme.

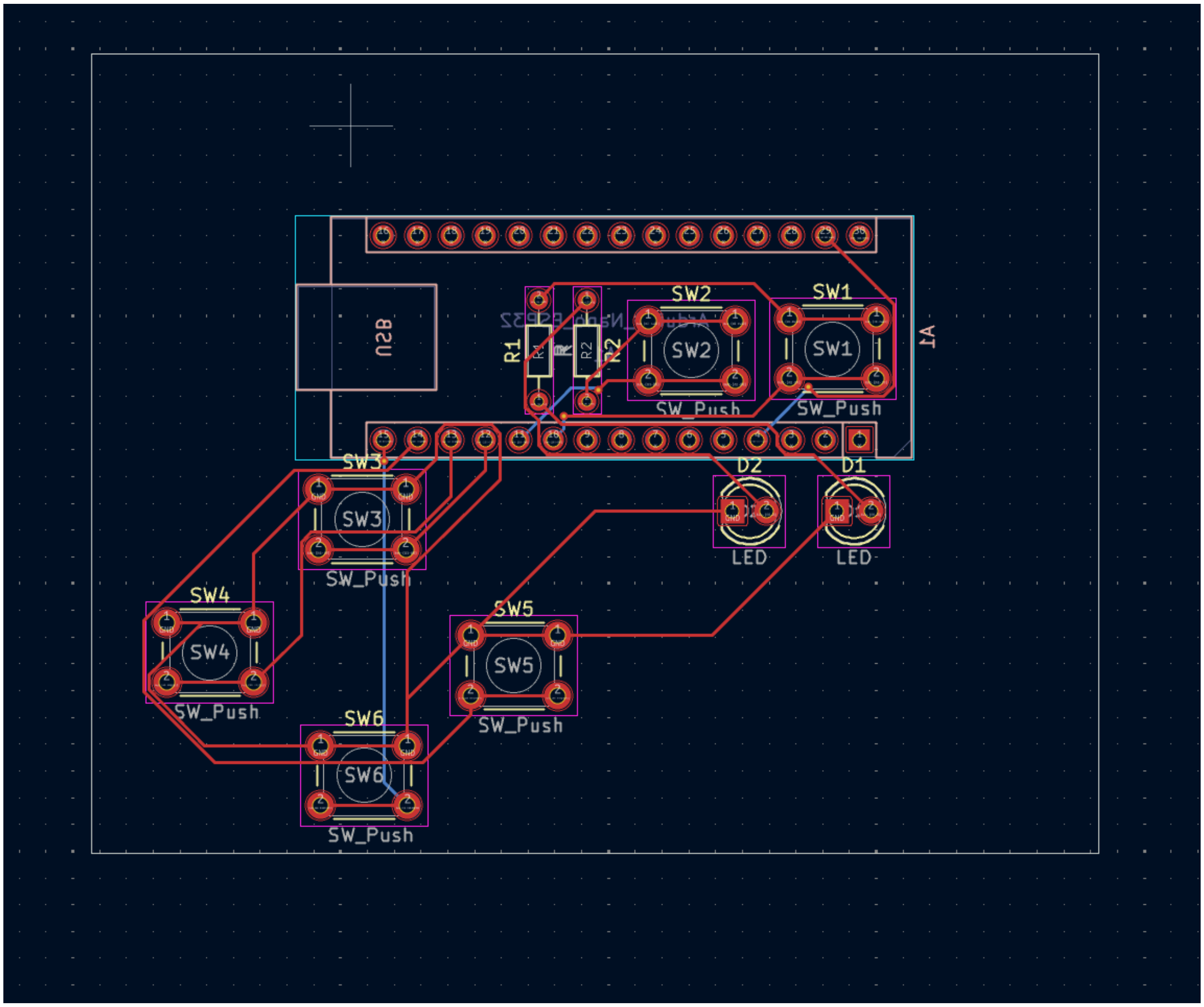
I made use of two LEDs on the buttons to make sure that the user would get visual feedback when any buttons were pressed. The A button would have a yellow led and the B button a Red led.

I could not sketch quickly without any apparatus other than my mouse, so I drew my sketch on the question paper. Please find it there.

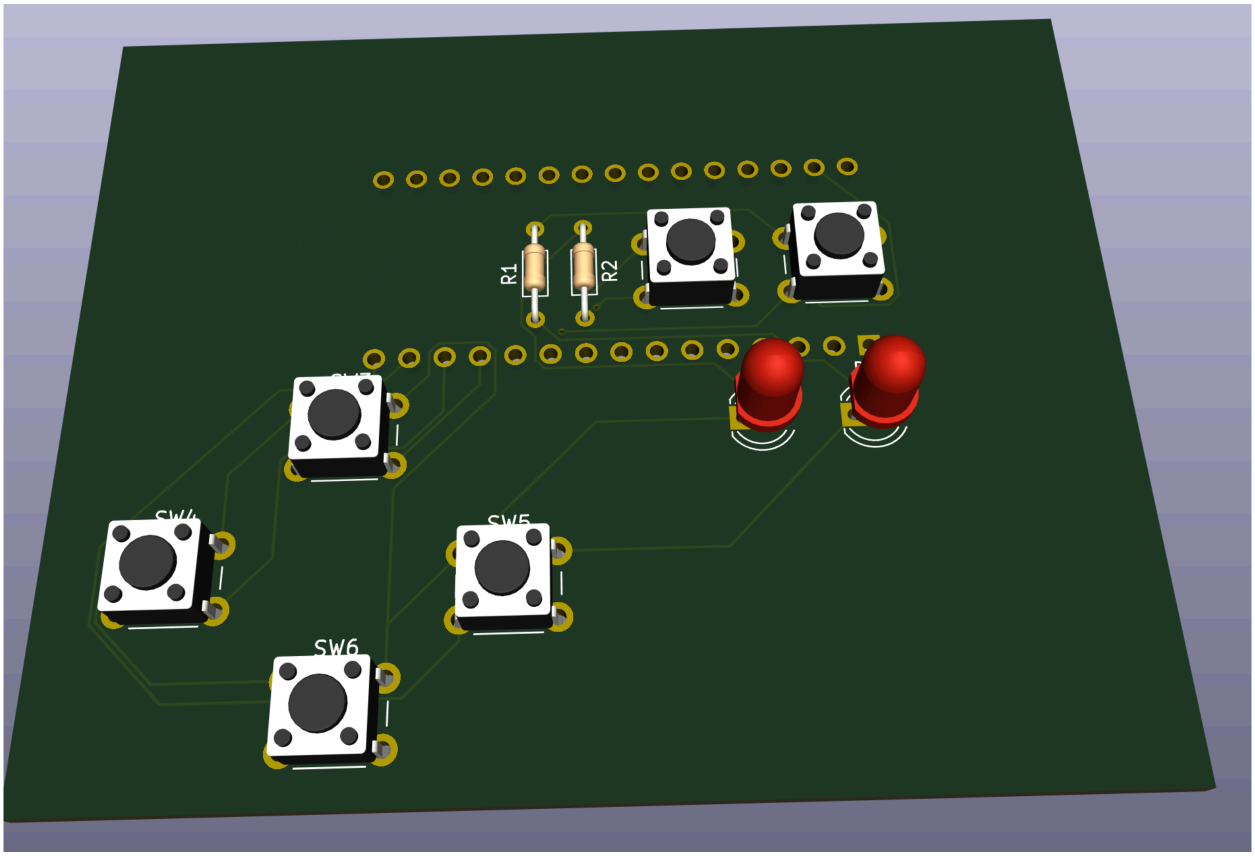
For the electronics I made use of KiCad, as despite the inability for it to show actual visual representation it’s electronic and design checker served to make sure that my design did not break any rules. I made use of the Arduino UNO microcontroller to allow for simple control of the functionality through code rather than pure electronic logic, and I connected the leds directly to the buttons to allow them to activate them without having to use the arduino. This allows for flexibility in how I program the arduino to react to buttons being pressed.



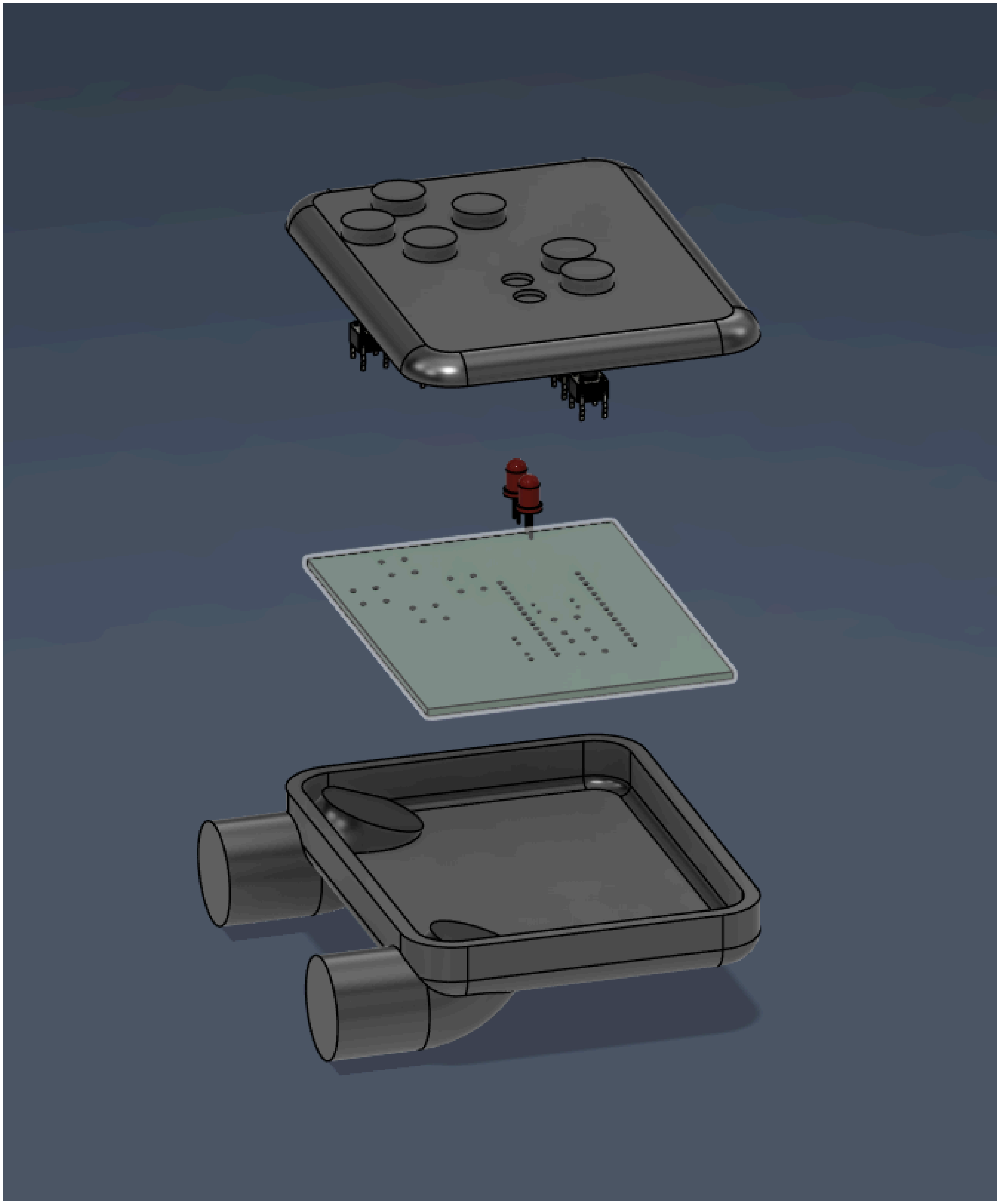
I also used it to design a quick PCB



Which was then used again in the design. I designed the PCB to meet the specific criteria of directional and A B buttons, as they are placed in an up down left right fashion and for the A and B Buttons, their corresponding LEDs are right below them. This allows the user to know exactly what button is being pressed and to receive a simple and effective visual response.



For the design, I used Fusion 360, and focused on making a simple yet comfortable fit for the components. I made use of various tools available to me, such as sketch to create the initial shapes, then extrusions and outlines to create the buttons, then I made use of revolve to help attach the cylinders to the final project.



This design allowed for a sensible and simple fit for the components as well as displaying the visual feedback on the LEDs.

This all culminated in the final design, which for the case of ExciteBike I believe represents a fun and interesting way to play the game.